

Team Number: _____ School & Team Name: _____

Student Names: _____

IMPOUND

1. T F 4.a. Team impounds one Vehicle (batteries stored separately), any spare parts or alignment devices, kit modification form (if used), and papers (if used). If a bottle pusher is used, it must be impounded and only 1 bottle pusher may be impounded.

CHECK-IN

2. T F 3.a.i No batteries containing lithium or lead acid are used. **If F, do not run and team receives participation points.**
3. T F 3.k. If kits are used, teams make at least two functional modifications to the kit during the build process that is documented on the Kit Modification Form. A bottle pusher from kit sources is not used.
4. _____ 4.c.iv. **Bottle Pusher Width:** The widest opening of the bottle pusher (cm). If team doesn't have a bottle pusher, do not fill out.

Construction Parameters:

- T F 4.c.iii.1 Team impounds up to 1 vehicle. **If F, all runs get a construction violation**
T F 3.a, 3.b Electrical Energy used by the Vehicle for any purpose solely comes from a maximum of 8 AA batteries and energy sources are in easily accessible locations for the ES. Propulsion energy is electric and comes from the AA batteries.
T F 3.a.ii Energy storage methods which allow teams to exceed the instantaneous energy available from the batteries are not used. (Ex: Large capacitors or a spring)
T F 3.a.iii Lasers are not used.
T F 3.d, 3.e Including all components, the vehicle length does not exceed 80.0 cm and the width does not exceed 35.0 cm at any point.
T F 3.f. A device to catch and push the Water Bottle is attached to the front of the Vehicle and is the foremost part of the Vehicle. The pusher is rigid, does not flex, change locations, have moving parts or the have the ability to change in size. The Pusher's foremost edge(s) has rounded edge(s) approximately the diameter of a #2 pencil or larger. The pusher can move bottles not perfectly round.
T F 3.g. The vehicle has a Measurement Point that is the rear most part of the vehicle.
T F 3.h. The vehicle is not remotely controlled or tethered.
T F 3.j. Team can answer questions on device design, construction, and operation per the Building Policy on www.soinc.org.
T F 3.i. Single-source programs are not used.

1st RUN 2nd RUN ← EVENT TIME (8 MINUTES FOR UP TO 2 RUNS)

Construction Parameters:

- T F T F 3.i. All parts of the Vehicle move as a whole, without any anchors, tethers, tie downs, launching ramps, or other separate pieces. The only parts that contact the floor during the run are the wheels/treads, and any other parts already in contact with the floor in the ready-to-run configuration.

5. T F 13. T F ALL CONSTRUCTION PARAMETERS ARE MET (Including Check-In Construction Parameters) (Recheck if necessary)

Competition Parameters:

- T F T F 4.c.v. Once participants start checking in, they do not leave or gain outside assistance or materials.
T F T F 4.c.vi. Participants do not use an AC outlet power during their time slot.
T F T F 4.c.vii. AI is not used during the day of the competition.
T F T F 4.d.ii. Sighting, aligning, measurement, and aiming devices are only used within the defined Track area, and devices are not secured to the track.
T F T F 4.d.iv. Any substances applied to the device are approved by the ES prior to use and do not damage or leave residue on the event area. The Track remains dry at all times. No tape is used on the event floor.
T F T F 4.d.v. Team only uses the internet to program their Device.
T F T F 4.d.vi. Team does not roll the Vehicle on the Track floor without tournament permission.
T F T F 4.d.vii.1 In ready-to-run configuration, the Vehicle's Measurement Point is over the Start Point within approx. 1 cm, the Vehicle's body is placed in front of the Start Point and is able to remain at the starting position without being touched until triggered. All items not part of the Vehicle are placed behind the Start Point before a run.
T F T F 4.d.vii.4 Participants use a #2 unsharpened pencil to start the Vehicle using a vertical motion and only the pencil contacts the Vehicle when triggered. Once the run starts and until the Vehicle is retrieved, competitors do not use, view, or access their programming tool(s).
T F T F 4.d.vii.9 Competitors do not follow their Vehicle down the track before called to retrieve their vehicle.

6. T F 14. T F ALL COMPETITION PARAMETERS ARE MET

7. **Success** 15. **Success** 8.q. **Successful Run:** Any run where the Vehicle moves before the end of Event Time and is NOT a failed run. A failed run includes the following: 1) The Vehicle travels the wrong direction from the Start Point. 2) The Vehicle starts before ES is ready. 3) The Vehicle's distance cannot be measured. 4) The team pushes the Vehicle down the track. 5) Any run that does not occur in the 8-minute Event Time.
Failed **Failed**

FILL OUT 8-12 IF RUN 1 IS SUCCESSFUL, AND 16-20 IF RUN 2 IS SUCCESSFUL.

8. _____ 16. _____ 5.d. **Run Time:** Time (sec.) from when the team actuates the Vehicle and ends when the vehicle comes to a complete stop. Run time is 0.00 sec. for failed runs.
9. _____ 17. _____ 5.c. **Vehicle Distance:** point-to-point measurement from the Vehicle Measurement Point to the Target Point, measured to the nearest 0.1 cm.
10. T F 18. T F 5.d.ii **Bottle Past Target:** The Water Bottle is past the target point and all of the following Moving Water Bottle Requirements are followed: 1) The Bottle remains in contact with the event floor at all times. 2) No part of the vehicle extends under the Bottle. 3) The Bottle remains upright at all times. 4) The Bottle may only be pushed. 5) Only the bottle pusher may contact the Water Bottle.
If this is T, continue to box 11 and 12, otherwise skip those boxes.
11. _____ 19. _____ 4.d.vi.7 **Bottle Distance:** Distance between the Water Bottle and the Bottle Line.
12. T F 20. T F 5.f. **Bottle Bonus:** The Water Bottle is fully beyond the Bottle Line.
21. _____ : _____ 8.n. **Event Time Left (National Only):** Event time ends after 2 runs or after the time limit has been reached. (min:sec)
22. T F General Rule: The team is disqualified. (Notify the team and their coach as soon as possible.)